Daedalus | Structures Lead

OBJECTIVE

Stable, recoverable rocket
4,000-ft design target

MY OWNERSHIP

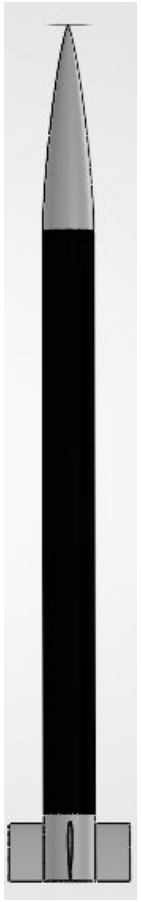
- Von Kármán nose + shoulder
- Fin can + structural integration
- Recovery attachment + interfaces
- PLA build, fit testing + validation

ENGINEERING CHALLENGES

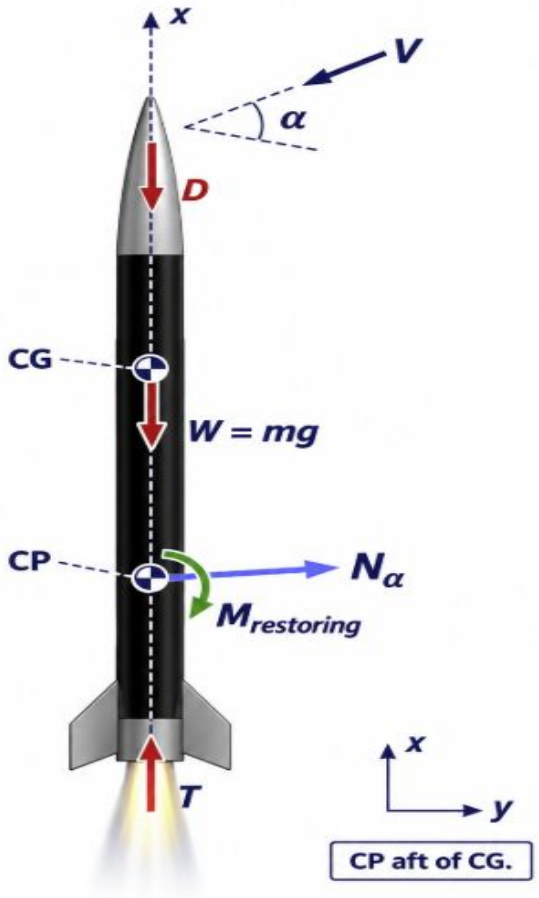
- Nose: drag vs. mass and printability
- Fin alignment after design review
- Simulation assumptions vs. testing

4,086 ft simulated **1.23 cal** modeled

SOLIDWORKS CAD

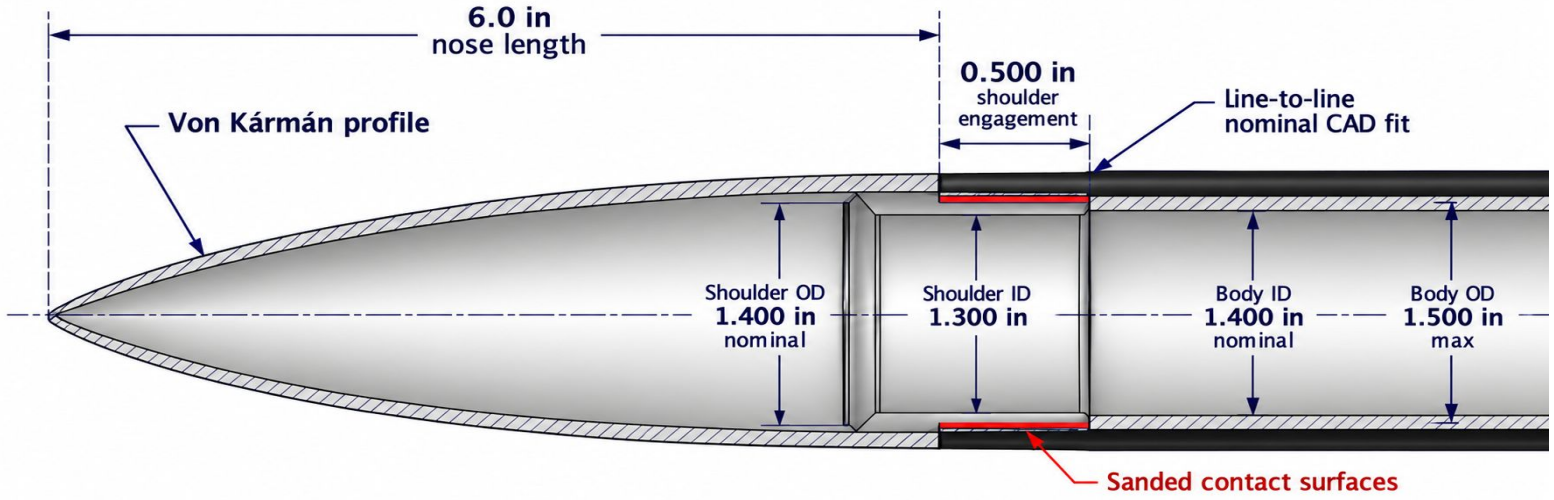


FORCES + STABILITY

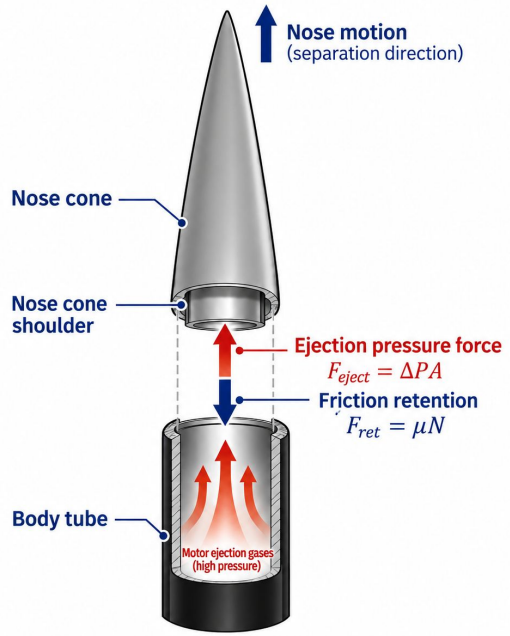


Built + ground-tested • never flown

Nose-Cone Design & Retention Testing



IDEALIZED NOSE SEPARATION FBD



PLA print → sand → spring scale

≈19 N extraction

$$F_{ret} \approx \mu N$$

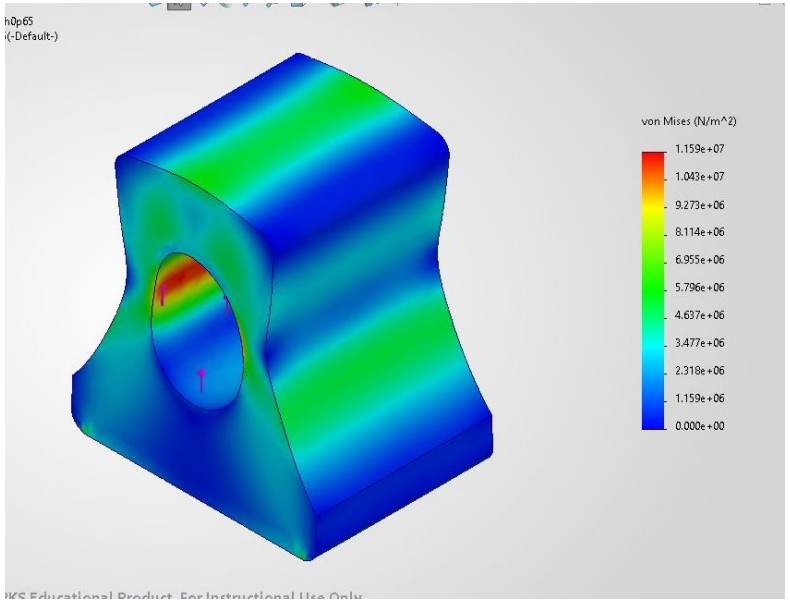
$$\Delta P_{min} = \frac{F_{pull}}{A} = \frac{19}{9.931 \times 10^{-4}} = 19.1 \text{ kPa} = 2.77 \text{ psi}$$

Gas-flow area = 1.539 − 1.343 = 0.196 in² (12.8% open), via proposed isolated 0.5-in av-bay bypass.

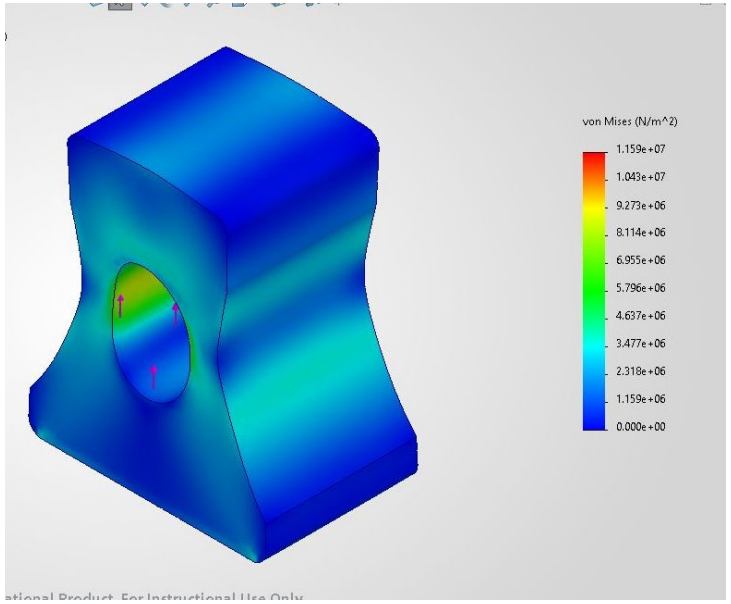
Motor → bypass → recovery cavity → nose. The 19.1 kPa threshold uses nose area; verify pressure rise and ejection.

Recovery Eyelet Design & Structural Analysis

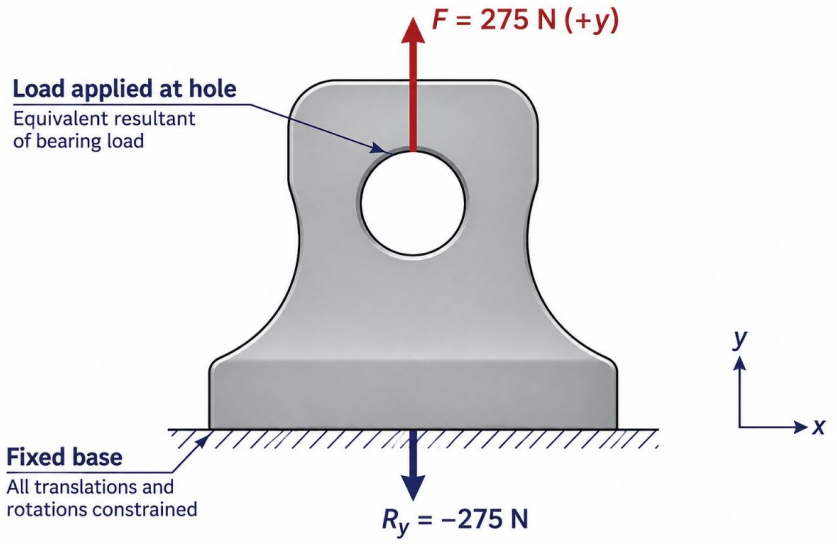
ORIGINAL-STYLE



WIDER PROFILE + HOLE LIGAMENT



FBD LOAD CASE



PEAK SIMULATED STRESS

−23.5% **11.591 → 8.869 MPa peak**

MAX DISPLACEMENT

−20.3% **0.04138 → 0.03299 mm**

Assumed 275 N • fixed root • isotropic ABS

Mesh Convergence Verification

- Four mesh sizes tested; refining from 0.80 to 0.65 mm changed peak stress by ≤1.6% and displacement by <0.2% for both designs.

KEY FINDINGS + LESSONS

- Wider, smoother profile: 23.5% lower peak stress
- Subsystem changes affect recovery and structure
- Analytical predictions need physical testing